

Short explanation of GSP and GSPy map scaling

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In DP (Design Point) simulation mode

- The maps are ignored for the performance calculations
 - W and Wc are determined by the exist conditions of the upstream component (e.g. inlet, combustor etc.)
 - PR, Eta are directly taken from the design values given in the design tab sheet.
 - (N and Nc are NOT used for performance calculation)
- Map scaling factors are calculated for subsequent OD simulations.
 - This is to allow for scaling of (similar) turbomachinery maps to a given cycle design point, especially in the conceptual cycle design phase or if actual maps are not available and maps of similar turbomachinery are used instead.
 - For map input
 - $SF_{Nc} = N_{cdes}/N_{cmapdes}$
 - Beta is not scaled ($Beta = Beta_{map}$)
 - $N_{cmapdes}$ and $Beta_{mapdes}$ determine the map design point. This is the scaling point corresponding to the cycle design data, indicated by the little yellow rectangle in GSP map plots.
 - For map output
 - $SF_{Wc} = W_{cdes}/W_{cmapdes}$
 - $SF_{PR} = (PR_{des}-1) / (PR_{mapdes}-1)$
 - $SF_{Eta} = Eta_{des}/Eta_{mapdes}$
 - *Wcmapdes, PRmapdes and Eta mapdes are read from the map using Ncmapdes and Beta mapdes*

In OD (Off-design) simulation mode

- Component performance is determined by state variables.
 - State variables generated during Newton-Raphson or similar solver iterations, determine component Nc and Beta values. In DP these state variables are 1. In OD mode they indicated where the component is operating relative to the design point by values deviating from 1. So $Nc = state_Nc * N_{cdes}$
 - Nc is scaled using SF_{Nc} to Ncmap by $N_{cmap} = Nc / SF_{Nc}$.
 - Now with $Beta_{map}$ (= Beta) and Ncmap known, the map table can be looked up to obtain Wcmap, PRmap and Eta map.
 - Next step is to scale back to the actual cycle design using
 - $Wc = W_{cdes} * SF_{Wc}$
 - $PR = (PR_{map} - 1) * SF_{PR} + 1$
 - $Eta = Eta_{map} * SF_{Eta}$
 - Now the rest of the component performance can be calculated and error variables (residuals) calculated for conservation of energy/power and mass flow (the deviation of incoming mass flow from the mass flow W according to the map operating point).

- With all operating conditions (inlet, fuel flow, deterioration, VSV angle etc.) unchanged from the DP operating conditions, the solution from the solver will exist of all state variable equal to 1 meaning the OD calculated operating point exactly matches the DP point.

OD mode with unscaled maps

- In case performance must be predicted with known maps, scaling for that map must be deactivated and the 'No scaling' option in the component window Map tab sheet must be checked.
- The 'No scaling' option is useful to predict component matching of a gas turbine system with fixed known turbomachinery designs.
- Now the maps will not be scaled to the DP data, meaning the DP cycle design is ignored and the OD simulation will iterate towards an operating point matching with the actual unscaled map(s). This OD point may well deviate from the DP point.
 - The DP mode simulation will only serve to calculate the scaling factors for determining how far the user specified design point is deviating from the operating point resulting from an OD simulation ('how far from 1.0').
 - The scaling factors can be selected in the output to be shown in the initial DP simulation results.
 - In the OD results, the scaling factors for the map output (W_c , PR and η_a) will all be 1.
 - The N_c scaling factor will remain as in DP to convert rpm to normalized map speed for example. This has no effect on the performance results.
 - *Note that with unscaled maps, the DP results only reflect the desired, expected or estimated design cycle performance. The DP simulation is ignoring the component maps, thereby also ignoring any component mismatch. The OD simulation results with unscaled maps show the actual (steady-state) operating point where the engine system is forced to stabilize given the fixed turbomachinery designs. This OD point will therefore often deviate from the DP point where component matching is ignored. The OD point also ignores the DP values for PR , W_c and η_a as it takes these values straight from the unscaled map as a function of actual N_c and β which both are determined by the state variables during the iteration.*
- To analyze effects of small changes in the turbomachinery design, such as NGV flow capacity for example, the scaling factor inputs in the Map tab sheet can be used.